

NUMB11: Numerical Linear Algebra: Assignment 1

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September 10, 2024

1. Task 1 solution:

A is a square matrix of dimensions $n \times n$. $\rho(A)$ is the spectral radius of A. We need to show that $\rho(A) \leq \|A\|$.

We know that

$$\rho(A) = \max_{1 \leq i \leq n} |\lambda_i(A)| \quad (1)$$

Or, the largest eigenvalue of matrix A.

If λ is the largest eigenvalue of matrix A, then there is a vector x such that

$$Ax = \lambda x \quad (2)$$

This is the eigenvector. Taking norm on both sides,

$$\|Ax\| = \|\lambda x\| \quad (3)$$

Since λ is a scalar,

$$\begin{aligned} \|Ax\| &= |\lambda| \|x\| \\ |\lambda| &= \frac{\|Ax\|}{\|x\|} \\ &\leq \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} \\ &= \|A\| \end{aligned} \quad (4)$$

So now we have got that $|\lambda| \leq \|A\|$ where $\rho(A) = \lambda$.

Hence,

$$\rho(A) \leq \|A\| \quad (5)$$

2. Task 2 solution:

A is a square matrix and 0 is the zero matrix of corresponding dimensions. We need to show that $\|A\| < 1 \implies \lim_{n \rightarrow \infty} A^n = 0$.

We know from Task 1 that $\rho(A) \leq \|A\|$. Now if $\|A\| < 1$, then $\rho(A) \leq \|A\| < 1$. Taking power of n on both sides,

$$\begin{aligned} (\rho(A))^n &\leq \|A\|^n \\ (\rho(A))^n &\leq \|A\|^n < 1 \end{aligned} \tag{6}$$

If λ is the largest eigenvalue of matrix A , then,

$$\begin{aligned} (|\lambda|)^n &< 1 \\ |\lambda| &< 1 \end{aligned} \tag{7}$$

This implies that $\lim_{n \rightarrow \infty} \lambda^n = 0$.

If λ is the largest eigenvalue of matrix A , then there is a vector x such that

$$\begin{aligned} Ax &= \lambda x \\ A^k x &= \lambda^k x \quad \text{for } k = 0, 1, 2, \dots \end{aligned} \tag{8}$$

So,

$$\begin{aligned} \lim_{n \rightarrow \infty} \lambda^n = 0 &\implies x \cdot \lim_{n \rightarrow \infty} \lambda^n = 0 \\ &\implies \lim_{n \rightarrow \infty} (\lambda^n x) = 0 \\ &\implies \lim_{n \rightarrow \infty} (A^n x) = 0 \\ &\implies \left(\lim_{n \rightarrow \infty} A^n \right) x = 0 \\ &\implies \lim_{n \rightarrow \infty} A^n = 0 \end{aligned} \tag{9}$$

Thus,

$$\|A\| < 1 \implies \lim_{n \rightarrow \infty} A^n = 0 \tag{10}$$

3. Task 3 solution:

3.1. $\|x\|_\infty \leq \|x\|_2$

We know that $\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$.

Let $1 \leq j \leq n$ such that $\max_{1 \leq i \leq n} |x_i| = |x_j|$.

So,

$$\|x\|_\infty = |x_j| \tag{11}$$

We know that

$$\|x\|_2 = \left(\sum_{i=1}^n |x_i|^2 \right)^{\frac{1}{2}} \tag{12}$$

For $1 \leq j \leq n$,

$$\begin{aligned} \left(\sum_{i=1}^n |x_i|^2 \right)^{\frac{1}{2}} &= \left(|x_j|^2 + \sum_{i=1, i \neq j}^n |x_i|^2 \right)^{\frac{1}{2}} \\ &\geq (|x_j|^2)^{\frac{1}{2}} \\ &= |x_j| = \|x\|_{\infty} \end{aligned} \tag{13}$$

From 11 and 13, we can conclude that,

$$\|x\|_{\infty} \leq \|x\|_2 \tag{14}$$

3.2. $\|x\|_2 \leq \sqrt{n}\|x\|_{\infty}$

We know that,

$$\|x\|_2 = \left(\sum_{i=1}^n |x_i|^2 \right)^{\frac{1}{2}} \tag{15}$$

We can rewrite this as,

$$\begin{aligned} \|x\|_2 &= \left(\sum_{i=1}^n |x_i|^2 \right)^{\frac{1}{2}} \\ &= (|x_1|^2 + |x_2|^2 + \dots + |x_n|^2)^{\frac{1}{2}} \\ &\leq \left(\max_{1 \leq i \leq n} |x_i|^2 + \max_{1 \leq i \leq n} |x_i|^2 + \dots + \max_{1 \leq i \leq n} |x_i|^2 \right)^{\frac{1}{2}} \\ &= \left(n \max_{1 \leq i \leq n} |x_i|^2 \right)^{\frac{1}{2}} \\ &= \sqrt{n} \max_{1 \leq i \leq n} |x_i|^2 \\ &= \sqrt{n} \|x\|_{\infty} \end{aligned} \tag{16}$$

Thus,

$$\|x\|_2 \leq \sqrt{n}\|x\|_{\infty} \tag{17}$$

3.3. $\|A\|_{\infty} \leq \sqrt{n}\|A\|_2$

From 3.1 we know that,

$$\begin{aligned} \|x\|_{\infty} &\leq \|x\|_2 \\ \implies \|Ax\|_{\infty} &\leq \|Ax\|_2 \end{aligned} \tag{18}$$

And, from 3.2 we know that,

$$\begin{aligned} \|x\|_2 &\leq \sqrt{n}\|x\|_{\infty} \\ \implies \frac{1}{\|x\|_2} &\geq \frac{1}{\sqrt{n}\|x\|_{\infty}}, \quad (x \neq 0) \end{aligned} \tag{19}$$

Using 18 and 19,

$$\begin{aligned} \frac{\|Ax\|_\infty}{\sqrt{n}\|x\|_\infty} &\leq \frac{\|Ax\|_2}{\|x\|_2}, \quad (x \neq 0) \\ \Rightarrow \frac{\|Ax\|_\infty}{\|x\|_\infty} &\leq \frac{\sqrt{n}\|Ax\|_2}{\|x\|_2}, \quad (x \neq 0) \end{aligned} \quad (20)$$

Taking supremum on both sides we finally get that,

$$\|A\|_\infty \leq \sqrt{n}\|A\|_2 \quad (21)$$

3.4. $\|A\|_2 \leq \sqrt{n}\|A\|_\infty$

From 3.1 we know that,

$$\begin{aligned} \|x\|_\infty &\leq \|x\|_2 \\ \Rightarrow \frac{1}{\|x\|_\infty} &\geq \frac{1}{\|x\|_2}, \quad (x \neq 0) \end{aligned} \quad (22)$$

And, from 3.2 we know that,

$$\begin{aligned} \|x\|_2 &\leq \sqrt{n}\|x\|_\infty \\ \Rightarrow \|Ax\|_2 &\leq \sqrt{n}\|Ax\|_\infty \end{aligned} \quad (23)$$

Using 22 and 23,

$$\frac{\sqrt{n}\|Ax\|_\infty}{\|x\|_\infty} \geq \frac{\|Ax\|_2}{\|x\|_2}, \quad (x \neq 0) \quad (24)$$

Taking supremum on both sides we finally get that,

$$\begin{aligned} \sqrt{n}\|A\|_\infty &\geq \|A\|_2 \\ \text{or, } \|A\|_2 &\leq \sqrt{n}\|A\|_\infty \end{aligned} \quad (25)$$

4. Task 4 solution:

A row matrix A and a vector a that have the same entries. That is,

$$A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} \quad \text{and} \quad a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad (26)$$

We know that,

$$\begin{aligned} \|A\|_2 &= \sqrt{\rho(A^T A)} \\ \text{and } \|a\|_2 &= \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2} \end{aligned} \quad (27)$$

Next we will try to find what $\rho(A^T A)$ is. First,

$$A^T = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad (28)$$

Hence,

$$A^T A = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} = \begin{bmatrix} a_1^2 & a_1 a_2 & \cdots & a_1 a_n \\ a_2 a_1 & a_2^2 & \cdots & a_2 a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n a_1 & a_n a_2 & \cdots & a_n^2 \end{bmatrix} \quad (29)$$

We get a $n \times n$ matrix. $A^T A$ is symmetric. It's eigenvalues are non-negative. In fact, it's only non-zero eigenvalue is, $a_1^2 + a_2^2 + \cdots + a_n^2$. Thus,

$$\rho(A^T A) = a_1^2 + a_2^2 + \cdots + a_n^2 \quad (30)$$

Therefore,

$$\begin{aligned} \|A\|_2 &= \sqrt{\rho(A^T A)} \\ &= \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2} \\ &= \|a\|_2 \end{aligned} \quad (31)$$

We have hence proved that $\|A\|_2 = \|a\|_2$.

Next, to find something similar for 1-norm and ∞ -norm.

We know that following,

$$\begin{aligned} \|a\|_1 &= |a_1| + |a_2| + \cdots + |a_n| \\ \|a\|_\infty &= \max_{1 \leq i \leq n} |a_i| \\ \|A\|_1 &= \max_{1 \leq j \leq n} \sum_{1 \leq i \leq n} |a_{ij}| \\ \|A\|_\infty &= \max_{1 \leq i \leq n} \sum_{1 \leq j \leq n} |a_{ij}| \end{aligned} \quad (32)$$

$\|A\|_1$ is basically finding the sum of the absolute values of each column and then taking the biggest value.

Or, for the given row matrix A,

$$\begin{aligned} \|A\|_1 &= \max_{1 \leq i \leq n} |a_i| \\ &= \|a\|_\infty \end{aligned} \quad (33)$$

And, $\|A\|_\infty$ is basically finding the sum of the absolute values of each row and then taking the biggest value.

Or, for the given row matrix A,

$$\begin{aligned}\|A\|_\infty &= \sum_{1 \leq i \leq n} |a_i| \\ &= \|a\|_1\end{aligned}\tag{34}$$

Thus, we have found that,

$$\begin{aligned}\|A\|_1 &= \|a\|_\infty \\ \text{and } \|A\|_\infty &= \|a\|_1\end{aligned}\tag{35}$$

5. Task 5 solution:

For any matrix A, the Frobenius norm is,

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}\tag{36}$$

where $a_{ij} \in A$ and $A \in \mathbb{R}^{m \times n}$

When multiplying matrix A by its transpose A^T , $A^T A$ is a square matrix. The element in the i^{th} row and i^{th} column of $A^T A$, or the diagonal elements is given by,

$$(A^T A)_{ii} = \sum_{k=1}^n a_{ki} \cdot a_{ki} = \sum_{k=1}^n a_{ki}^2\tag{37}$$

Trace($A^T A$) is the sum of these diagonal elements.

$$\text{trace}(A^T A) = \sum_{i=1}^n (A^T A)_{ii} = \sum_{i=1}^n \sum_{k=1}^n a_{ki}^2\tag{38}$$

From 36 and 38, we can conclude that,

$$\|A\|_F = \sqrt{\text{trace}(A^T A)}\tag{39}$$

Next, we have that Q is an orthogonal matrix with suitable dimensions. It has a property $Q^T Q = I$ where I is the identity matrix. Now, using the previous result,

$$\begin{aligned}\|QA\|_F &= \sqrt{\text{trace}((QA)^T(QA))} \\ &= \sqrt{\text{trace}((A^T Q^T)(QA))} \\ &= \sqrt{\text{trace}(A^T Q^T Q A)} \quad (Q^T Q = I) \\ &= \sqrt{\text{trace}(A^T A)} \\ &= \|A\|_F\end{aligned}\tag{40}$$

Thus, $\|QA\|_F = \|A\|_F$

6. Task 6 solution:

$\mathbf{u}, \mathbf{v}$ are 2 vectors of n dimension. So,

$$\mathbf{u}\mathbf{v}^T = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \quad (41)$$

We know that rank of a matrix A is the dimension of the vector space spanned (or generated) by its columns (or rows). That is, it's the maximal number of linearly independent columns (or rows). Now,

$$\mathbf{u}\mathbf{v}^T = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} = \begin{bmatrix} u_1 v_1 & u_1 v_2 & \cdots & u_1 v_n \\ u_2 v_1 & u_2 v_2 & \cdots & u_2 v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_n v_1 & u_n v_2 & \cdots & u_n v_n \end{bmatrix} \quad (42)$$

Columns of the outer product are all proportional to the first column. Therefore, they are all linearly dependent on that one column.

Or we can say that, all columns (or rows) of a matrix are scalar multiples of each other or, that each row is a multiple of $\mathbf{v}$ and each column is a multiple of $\mathbf{u}$.

This means that the matrix can be described by a single row or column. Which means that only one column (or row) is needed to describe the entire matrix.

Thus, we can say that they are linearly dependent. And there is only one column (or row) that is linearly independent, one independent vector. (The rest being scalar multiples of this).

Hence, the outer product $\mathbf{u}\mathbf{v}^T$ is a $n \times n$ matrix of rank 1.

Next, computing the 2-norm of the outer product, $\mathbf{u}\mathbf{v}^T = A$

$$\begin{aligned} \|A\|_2 &= \sqrt{\rho((A)^T(A))} \\ &= \sqrt{\rho((\mathbf{u}\mathbf{v}^T)^T(\mathbf{u}\mathbf{v}^T))} \\ &= \sqrt{\rho(\mathbf{v}\mathbf{u}^T\mathbf{u}\mathbf{v}^T)} \\ &= \sqrt{\mathbf{u}^T\mathbf{u}}\sqrt{\rho(\mathbf{v}\mathbf{v}^T)} \quad \left(\rho(\mathbf{u}^T\mathbf{u}) = \mathbf{u}^T\mathbf{u} = \sum_{i=1}^n u_i^2, \text{ a scalar value} \right) \end{aligned} \quad (43)$$

$\mathbf{v}\mathbf{v}^T$ is a square matrix with rank 1. It has eigenvalues 0 and the trace $= \sum_{i=1}^n v_i^2$. Thus, the largest eigenvalue is $\sum_{i=1}^n v_i^2$.

$$\begin{aligned}
&= \sqrt{\left(\sum_{i=1}^n u_i^2\right)\left(\sum_{i=1}^n v_i^2\right)} \\
&= \|u\|_2 \|v\|_2
\end{aligned} \tag{44}$$

Thus, the 2-norm of the outer product is $\|u\|_2 \|v\|_2$.

7. Task 7 solution:

$\mathbf{u}\mathbf{v}^T$ is the outer product of two vectors. We need to find it's singular value decomposition (SVD) in terms of these two vectors.

SVD of a matrix $A = U\Sigma V^T$ where,

$U = m \times m$ orthogonal matrix

$\Sigma = m \times n$ diagonal matrix with non-negative real values in the diagonal (singular values)

$V^T = n \times n$ orthogonal matrix V 's transpose

There are vectors $\mathbf{u}_1, \dots, \mathbf{u}_m$ such that,

$$U = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \dots \quad \mathbf{u}_m] \tag{45}$$

And there are vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ such that,

$$V^T = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_n^T \end{bmatrix} \tag{46}$$

$$\text{Let } \Sigma = \begin{bmatrix} \sigma_1 & \dots & 0 & \\ \vdots & \ddots & \vdots & O \\ 0 & \dots & \sigma_r & \\ & O & & O \end{bmatrix}$$

Then,

$$\begin{aligned}
A &= U\Sigma V^T \\
&= [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \dots \quad \mathbf{u}_m] \begin{bmatrix} \sigma_1 & \dots & 0 & \\ \vdots & \ddots & \vdots & O \\ 0 & \dots & \sigma_r & \\ & O & & O \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_n^T \end{bmatrix}
\end{aligned} \tag{47}$$

$$\begin{aligned}
&= [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_r \quad \mathbf{u}_{r+1} \quad \cdots \quad \mathbf{u}_m] \begin{bmatrix} \sigma_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_r \\ & O & O \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \vdots \\ \mathbf{v}_r^T \\ \mathbf{v}_{r+1}^T \\ \vdots \\ \mathbf{v}_n^T \end{bmatrix} \\
&= [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_r] \begin{bmatrix} \sigma_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_r \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \vdots \\ \mathbf{v}_r^T \end{bmatrix} + [\mathbf{u}_{r+1} \quad \cdots \quad \mathbf{u}_m] [O] \begin{bmatrix} \mathbf{v}_{r+1}^T \\ \vdots \\ \mathbf{v}_n^T \end{bmatrix} \quad (48) \\
&= [\sigma_1 \mathbf{u}_1 \quad \cdots \quad \sigma_r \mathbf{u}_r] \begin{bmatrix} \mathbf{v}_1^T \\ \vdots \\ \mathbf{v}_r^T \end{bmatrix} \\
&= \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T \\
&= \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T
\end{aligned}$$

Thus,

$$A = U \Sigma V^T = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T \quad (49)$$

where r is the rank of the matrix A and $\mathbf{u}_i \mathbf{v}_i^T$ are the outer products.

Next, let's provide a numeric example: Let $A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$

Let's begin with finding U .

$$A^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix}$$

Next, we have to find the eigenvalues and corresponding eigenvectors of AA^T . We know that eigenvectors are defined by the equation $\vec{A}v = \vec{\lambda}v$, and applying this to AA^T gives us

$$\begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (50)$$

Rewriting it as,

$$\begin{aligned}
11x_1 + x_2 &= \lambda x_1 \\
x_1 + 11x_2 &= \lambda x_2
\end{aligned} \quad (51)$$

$$\begin{aligned}\text{Or } (11 - \lambda)x_1 + x_2 &= 0 \\ x_1 + (11 - \lambda)x_2 &= 0\end{aligned}$$

Solve for λ by taking the determinant equal to zero.

$$\begin{aligned}\begin{vmatrix} (11 - \lambda) & 1 \\ 1 & (11 - \lambda) \end{vmatrix} &= 0 \\ (11 - \lambda)(11 - \lambda) - 1 &= 0 \\ (\lambda - 10)(\lambda - 12) &= 0 \\ \lambda &= 10, \lambda = 12\end{aligned}\tag{52}$$

This gives us two eigenvalues $\lambda = 10, 12$. Plugging λ back into the original equations gives us our eigenvectors. For $\lambda = 10$, we get eigenvector $\begin{bmatrix} 1 & -1 \end{bmatrix}$. And, for $\lambda = 12$, we get eigenvector $\begin{bmatrix} 1 & 1 \end{bmatrix}$

We can write these eigenvectors as a matrix from the largest to smallest eigenvalue's eigenvector from the first column on the left towards the right.

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\tag{53}$$

We can convert this matrix into an orthogonal matrix by applying Gram-Schmidt orthonormalization. First, we'll normalize $\vec{v}_1$.

$$\vec{u}_1 = \frac{\vec{v}_1}{|\vec{v}_1|} = \frac{\begin{bmatrix} 1, 1 \end{bmatrix}}{\sqrt{1^2 + 1^2}} = \frac{\begin{bmatrix} 1, 1 \end{bmatrix}}{\sqrt{2}} = \begin{bmatrix} \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \end{bmatrix}\tag{54}$$

Compute

$$\begin{aligned}\vec{w}_2 &= \vec{v}_2 - \vec{u}_1 \cdot \vec{v}_2 * \vec{u}_1 \\ &= \begin{bmatrix} 1, -1 \end{bmatrix} - \begin{bmatrix} \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \end{bmatrix} \cdot \begin{bmatrix} 1, -1 \end{bmatrix} * \begin{bmatrix} \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \end{bmatrix} \\ &= \begin{bmatrix} 1, -1 \end{bmatrix} - 0 * \begin{bmatrix} \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \end{bmatrix} \\ &= \begin{bmatrix} 1, -1 \end{bmatrix} - \begin{bmatrix} 0, 0 \end{bmatrix} \\ &= \begin{bmatrix} 1, -1 \end{bmatrix}\end{aligned}\tag{55}$$

and normalizing to give U .

$$\vec{u}_2 = \frac{\vec{w}_2}{|\vec{w}_2|} = \begin{bmatrix} \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \end{bmatrix}\tag{56}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}\tag{57}$$

Next, we will find out what V is. V is similarly calculated based on $A^T A$.

$$A^T A = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{bmatrix}\tag{58}$$

In the same way we did for U , we find the eigenvalues of matrix $A^T A$ as $\lambda_1 = 0, \lambda_2 = 10$ and $\lambda_3 = 12$ using $(A^T A) - \lambda I = 0$. And we find the corresponding eigenvectors as $\vec{v}_1 = [1, 2, -5]$, $\vec{v}_2 = [2, -1, 0]$ and $\vec{v}_3 = [1, 2, 1]$ respectively using $(A^T A)\vec{v} = \lambda\vec{v}$.

We will order $\vec{v}_1$, $\vec{v}_2$ and $\vec{v}_3$ as column vectors in a matrix according to size of the eigenvalue to get,

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & -5 \end{bmatrix} \quad (59)$$

We can convert this matrix into an orthogonal matrix by applying Gram-Schmidt orthonormalization.

$$\begin{aligned} \vec{u}_1 &= \frac{\vec{v}_1}{|\vec{v}_1|} = \left[\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right] \\ \vec{w}_2 &= \vec{v}_2 - \vec{u}_1 \cdot \vec{v}_2 * \vec{u}_1 = [2, -1, 0] \\ \vec{u}_2 &= \frac{\vec{w}_2}{|\vec{w}_2|} = \left[\frac{2}{\sqrt{5}}, \frac{-1}{\sqrt{5}}, 0 \right] \\ \vec{w}_3 &= \vec{v}_3 - \vec{u}_1 \cdot \vec{v}_3 * \vec{u}_1 - \vec{u}_2 \cdot \vec{v}_3 * \vec{u}_2 = \left[\frac{-2}{3}, \frac{-4}{3}, \frac{10}{3} \right] \\ \vec{u}_3 &= \frac{\vec{w}_3}{|\vec{w}_3|} = \left[\frac{1}{\sqrt{30}}, \frac{2}{\sqrt{30}}, \frac{-5}{\sqrt{30}} \right] \end{aligned} \quad (60)$$

This would give,

$$V = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} & 0 & \frac{-5}{\sqrt{30}} \end{bmatrix} \quad (61)$$

Thus,

$$V^T = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix} \quad (62)$$

For Σ , we take square roots of the non-zero eigenvalues and write them as the diagonals putting the largest in the first column, next largest in the next column and so on until the smallest. The diagonal entries are the singular values of A . Hence,

$$\Sigma = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix} \quad (63)$$

Thus, we finally have,

$$A = U\Sigma V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix} \quad (64)$$

Verifying we get,

$$A = U\Sigma V^T = \begin{bmatrix} \frac{\sqrt{12}}{\sqrt{2}} & \frac{\sqrt{10}}{\sqrt{2}} & 0 \\ \frac{\sqrt{12}}{\sqrt{2}} & -\frac{\sqrt{10}}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix} = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \quad (65)$$

This gives the same solution as $\sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T = A$.

We can verify 64 and 65 using Python programming. The code and it's output is given in Appendix 1.

1 Appendix

Python Code for Task 7:

```
import numpy as np
import scipy.linalg as sl

m = 2; n = 3

# Define the matrix A
A = np.array([[3, 1, 1], [-1, 3, 1]])

# Perform Singular Value Decomposition using scipy.linalg
U, s, V = sl.svd(A)
Sigma = sl.diagsvd(s,m,n)

# U is the left singular vectors
print("U:", U)

# s is the singular values.
# Sigma is the diagonal matrix with the singular values
print("Sigma:", Sigma)

# Vt is the transpose of the right singular vectors
print("Vt:", V)
```

Output:

```
U: [[-0.70710678 -0.70710678]
     [-0.70710678  0.70710678]]
Sigma: [[3.46410162 0.          0.          ]
         [0.          3.16227766 0.          ]]
Vt: [[-4.08248290e-01 -8.16496581e-01 -4.08248290e-01]
      [-8.94427191e-01  4.47213595e-01  5.26260748e-16]
      [-1.82574186e-01 -3.65148372e-01  9.12870929e-01]]
```